Dean Menezes

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Experience

University of Texas at Austin – Applied Research Laboratories (ARL)
Engineering Scientist *Dec 2025–Present*

Corpus Christi Independent School District (CCISD)
Programmer / Analyst *Aug 2025–Nov 2025*

Fortanix
Software Engineer *Mar 2022–Mar 2025*

- Worked on cryptography and cybersecurity software based on Intel SGX (Software Guard Extensions), enabling secure enclave-based execution.
- Implemented frontend and backend software for key management and secure processing of machine learning datasets using C++ and Rust.
- Led design and implementation of AWS Nitro Enclave File Persistence for the Confidential Computing Manager (CCM), enabling encrypted file persistence across enclave sessions.
- Developed SQL query and join logic components for the Data Execution Engine, supporting integration testing and production customer workloads.
- Contributed to Confidential AI (CAI) system improvements for secure execution of containerized Python-based AI workloads inside Intel SGX enclaves.

Google
Software Engineer Intern *2022*

- Developed C++ software for record-replay testing of derived intent generation used in Search infobox responses (weather, sports, finance).
- Improved debugging infrastructure and test coverage of production intent logic.
- Collaborated cross-functionally with Search teams to integrate testing tools into CI pipelines.

Snowflake
Software Engineering Intern *Jun 2021–Dec 2021*

- Implemented SQL compiler optimizations (filter pushdown, JOIN elimination) in Java, reducing query execution time by approximately 10% for common workloads.
- Enabled Bloom filter propagation across distributed query plan splits, improving selectivity and reducing I/O and memory usage, yielding up to 20% faster execution in benchmarks.
- Improved cardinality estimation accuracy for complex SQL queries, reducing suboptimal query plans by up to 20% in test cases.

Education

The University of Texas at Austin
M.S. in Computer Science, GPA: 4.00 *Spring 2025–Present*

University of California, Los Angeles
M.A. in Mathematics, GPA: 3.99 *2017–2021*

- Qualifying Exams Completed: Algebra, Logic, and Analysis
- University-wide highest grades on Algebra, Logic, and Analysis qualifying exams (Spring 2019)
- Relevant Coursework: Combinatorics, Model Theory, Set Theory, Number Theory, Topology

The University of Texas at Austin
B.S. Mathematics; B.A. Plan II Honors, Highest Honors, GPA: 3.98 *2012–2016*

- **Undergraduate Thesis:** Cartesian Closed Categories and Typed Lambda Calculus
- **Budapest Semesters in Mathematics (Fall 2014):** High Honors

Society of Actuaries
Actuarial Examinations *2025*

- Exam P/1 (Probability) — Passed September 2025
- Exam FM/2 (Financial Mathematics) — Passed October 2025

Technical Skills

Languages: C, C++, Python, Java, Rust, SQL
Technologies/Frameworks: NumPy, Pandas, SciPy, Scikit-Learn, FastAPI, BeautifulSoup, PyMongo, Matplotlib, Seaborn
Developer Tools: Linux/Unix, Wireshark, ANTLR, Flex/Yacc, Docker